

CSCE 633: Machine Learning

Perceptron: Multilayer Perceptrons

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Perceptron: Recap

- As we saw earlier, a perceptron can be written as:

$$h(x) = \text{sign} \left(\sum_{i=1}^d w_i x_i + w_0 \right)$$

or

$$h(x) = \text{sign} (\mathbf{w}^T \mathbf{x})$$

where $\mathbf{w} = [w_0, w_1, \dots, w_d]^T$ and $\mathbf{x} = [1, x_1, \dots, x_d]^T$

- Note: the bias term w_0 lumped with coefficients w_i for $i \in \{1, 2, \dots, d\}$ and added a corresponding dimension with value 1 to the input feature vector $\mathbf{x}$.

Multilayer Perceptron: Motivation

- Single layer perceptron:

$$h(x) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

- A major restriction:
 - Decision boundaries are linear
 - A huge limitation in what perceptrons can represent
 - Example: XOR problem

XOR Problem

- Consider using a perceptron for learning the exclusive OR (XOR) function:

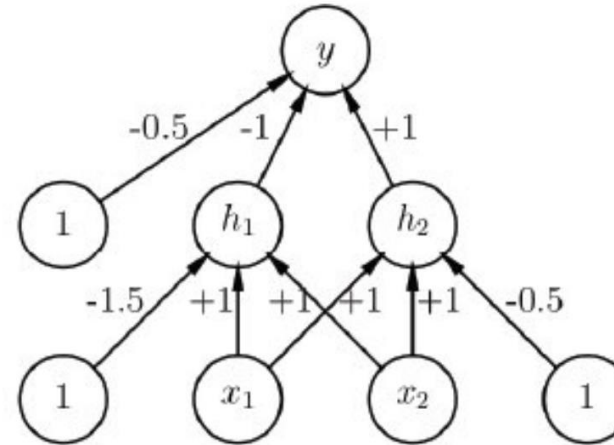
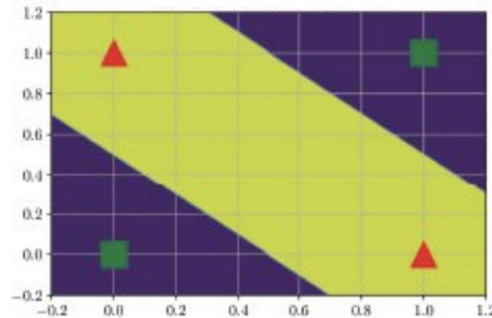
x1	x2	x1 XOR x2
0	0	0
0	1	1
1	0	1
1	1	0



- It is **NOT** linearly separable, so a perceptron cannot represent it.

XOR Problem: Solution

- Solution to XOR problem: stacking multiple perceptrons



- And XOR can be learned:
 - $h_1 = x_1 \wedge x_2$
 - $h_2 = x_1 \vee x_2$
 - $y = \neg h_1 \wedge h_2$
- Nodes h_1 and h_2 are called *hidden units*.
- This is called multi-layer perceptron.

Conclusion

- A single layer perceptron is limited to linearly separable data.
- Stacking perceptrons increases their flexibility.
- Can be shown: MLP can represent any logical function.